

A mini autonomous boat for a dad-and-son build

Concept Selection Report: options considered, the recommended concept, and what we need to decide before architecture design.

Recommendation: build **Concept G, a modular 3D-printed GPS catamaran**. It has two brains: an off-the-shelf **ArduPilot** autopilot that enforces a geofence and brings the boat home, and a small **camera 'mission brain'** that takes the photos. Instructions like "explore the pond, photograph some ducks, then come back" are turned into a checked mission on the laptop before launch. Estimated cash spend **≈£105** (≈£90 if you already own a 3S battery).

1. Summary

We want a small boat that a child can help design and build, that can be sent off with a plain-English instruction, and that **will not be lost in the lake**. The target budget is £100. Seven concepts were scored against cost, fun, intelligence and reliability. Reliability was weighted highest because it is the only hard requirement.

Concept G, the modular GPS catamaran, scored 4.45/5. The next best was 3.70. G stays the winner if any single criterion's weight is doubled. It wins because it separates the clever part from the safe part:

- **Safety brain.** ArduPilot Rover, open-source autopilot firmware used on thousands of boats and rovers. Geofence, return-to-launch, battery and lost-link failsafes, and stuck detection are mature, configurable features, so we don't have to write them.
- **Mission brain.** A £14 ESP32-S3 camera board acts as the Wi-Fi link and takes photos. It can only ask the autopilot to do things, and the autopilot still enforces the fence. Bugs in our own code cannot take the boat out of the pond.
- **Modular mechanical build.** Printed hull segments, clip-on thruster pods, a mast kit and an M3-grid 'Meccano rail' deck. Your son can design and swap his own modules.
- **Unsinkable by construction.** Foam-filled hulls float even if everything leaks, and a catamaran is very hard to capsize.

Key figure	Estimate	Basis
Size	≈600 × 340 mm, mast ≈300 mm	Fits a car boot; hull segments ≤200 mm fit a typical print bed
Mass	≈1.4 kg	Section 5.7 weight budget
Cruise speed	≈1 m/s (walking pace)	Slow is quieter, safer and gives sharper photos
Endurance	≥60 min (to be proven)	≈26 Wh usable, ≈10 W cruise; missions planned at ≤20 min
Link range	≈100-200 m over water (Wi-Fi)	Link not needed for safety: failsafes live on the boat
Cash spend	≈£105	Section 5.6; filament and fasteners from workshop stock

2. What 'good' looks like

The brief turned into criteria, each with a measurable test and a weight for scoring:

Criterion	Weight	What we are really asking	Pass test
Reliability (can't lose it)	30%	Does the boat come back, or stay findable and reachable, whatever fails: link, battery, GPS, software, weed, wind?	No single failure leaves the boat unrecoverable. Every failure mode ends in 'returns', 'holds inside the fence' or 'drifts to our bank'.
Cost	25%	Around £100 cash for a working Mk1. Upgrades can come later.	Core bill of materials $\leq$ ≈£105 at current UK/AliExpress prices.
Fun (build)	25%	Is there a real, modular mechanical build a child can own parts of, and keep modifying?	At least 4 swappable modules; child-designed parts possible with Tinkercad-level CAD.
Intelligence	20%	Can we say "explore this pond, take pictures of ducks, come back" and have it happen?	Plain-English request becomes a validated mission, runs hands-off, and returns tagged duck photos.

Assumptions

- Calm inland fresh water: a park pond or small lake, roughly 50-300 m across. Not rivers with current, not the sea.
- Operated from the bank by an adult, with the child as co-pilot. Nobody enters the water to recover it.
- 3D printer (PETG), soldering and general workshop tools are available, and software/electronics are within your comfort zone.
- Prices are September 2026 estimates from typical UK and AliExpress sellers. Treat them as $\pm 20\%$.

3. Concepts considered

Seven genuinely different ways to meet the brief. Each was scored 1 (poor) to 5 (excellent) per criterion.

	Concept	How it works	Strengths	Weaknesses
A	Buy a GPS bait boat and hack it	Commercial carp-fishing bait boat (already has GPS return-home); add a camera and try to script it.	Proven hull, sealed, built-in return-home.	£150-300 before mods; closed firmware; little for your son to build.
B	Tethered 'dog-on-a-lead' boat	Simple RC/auto boat on a 30-50 m floating line reeled from the bank.	Cheapest; can literally always be pulled back.	Tether snags reeds/weed and ducks; 'explore' limited to a circle; little autonomy to learn.
C	Scratch-built Pi/ESP32 boat, our own autopilot	Raspberry Pi or ESP32 with GPS, write navigation and failsafes ourselves.	Cheap, maximum learning, total control.	Every failsafe is our own untested code - the riskiest way to meet 'never lose it'.
D	Airboat (air propellers) on ArduPilot	Flat-bottom hull, two above-water fans, skid steering.	Weed-proof, no underwater seals, simple.	Loud (scares the ducks we want to photograph), blown about by wind, exposed spinning props near small fingers.
E	Paddle-wheel steamer style on ArduPilot	Twin side paddle wheels driven by geared DC motors.	Hugely fun mechanically, weed-tolerant, charming.	Slow, heavy, big to print; poor into wind; gearbox wear.
F	Convert a cheap RC toy boat	£30 RC speedboat with a flight controller squeezed inside.	Quick start, ready-made hull.	Tiny, not sealed, too fast, no room for camera/batteries; nothing modular to build.
G	Modular 3D-printed catamaran, ArduPilot + camera 'mission brain'	Foam-filled printed twin hulls, clip-on thruster pods, ArduPilot Rover on a cheap flight controller, ESP32-S3 camera as Wi-Fi/photo companion.	Proven failsafes & geofence, fully modular build, quiet, stable camera platform, clear upgrade path.	Tight on £100; weed around submerged props needs a guard; more build effort.

Also ruled out early: an aerial drone (different regulations, and a crash over water means a lost drone) and a submarine (radio and GPS don't work underwater, which conflicts with the 'can't lose it' requirement).

4. Evaluation

Concept	Cost	Fun	Intel.	Reliab.	Weighted
G Modular GPS catamaran	4	5	5	4	4.45
D ArduPilot airboat	4	4	4	3	3.70
C Scratch-built autopilot	5	4	4	2	3.65
E Paddle-wheel explorer	3	5	3	3	3.50
F Converted RC toy boat	4	2	4	3	3.20
B Tethered boat	5	2	1	4	3.15
A Hacked GPS bait boat	1	2	3	5	2.85
Weight	25%	25%	20%	30%	

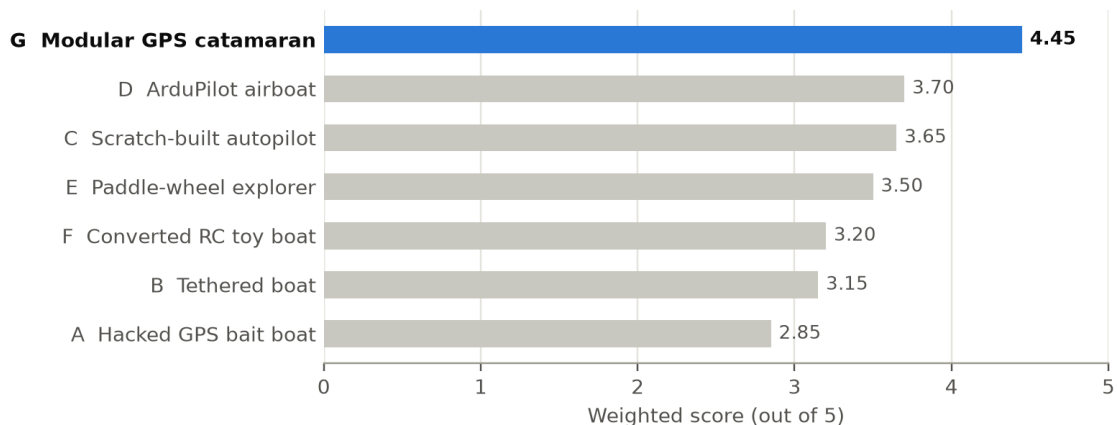


Figure 1. Weighted scores. Concept G leads the runner-up by 0.75 points.

Why G wins, and what nearly beat it

- **vs C (scratch-built autopilot):** the same fun and intelligence at a similar price, but C makes us write and prove every failsafe ourselves. G gives the learning at the mission level and borrows the safety level from a mature project. You can still write your own navigation later as an ArduPilot Lua script or companion code, with the fence still underneath it.
- **vs D (airboat):** D is weed-proof, which matters on ponds. But a fan is loud and the brief is to photograph ducks. G takes the good idea anyway: the airboat is offered as a swap-in thruster module for weedy water.
- **vs A (bait boat):** the most reliable out of the box, but over budget and closed. There's almost nothing for a young engineer to build.
- **vs B (tether):** it feels safe, but the line snags reeds, weed and wildlife, and a snagged tether is exactly how you end up losing a boat.

Sensitivity: doubling any one weight (then re-normalising) leaves the winner unchanged (cost: G, fun: G, intel: G, rel: G).

5. The recommended concept: Boaty Mk1

5.1 Overview

A 600 mm catamaran. It has two foam-filled printed hulls joined by two crossbeams that carry a sealed electronics box and a mast. Two clip-on thruster pods at the stern steer by differential thrust (no rudder), so it can turn on the spot and reverse, which also helps shake off weed. A forward-looking camera sits on the front of the box. The GPS and compass sit up the mast, away from motor currents. The GPS and compass sit up the mast, away from motor currents.

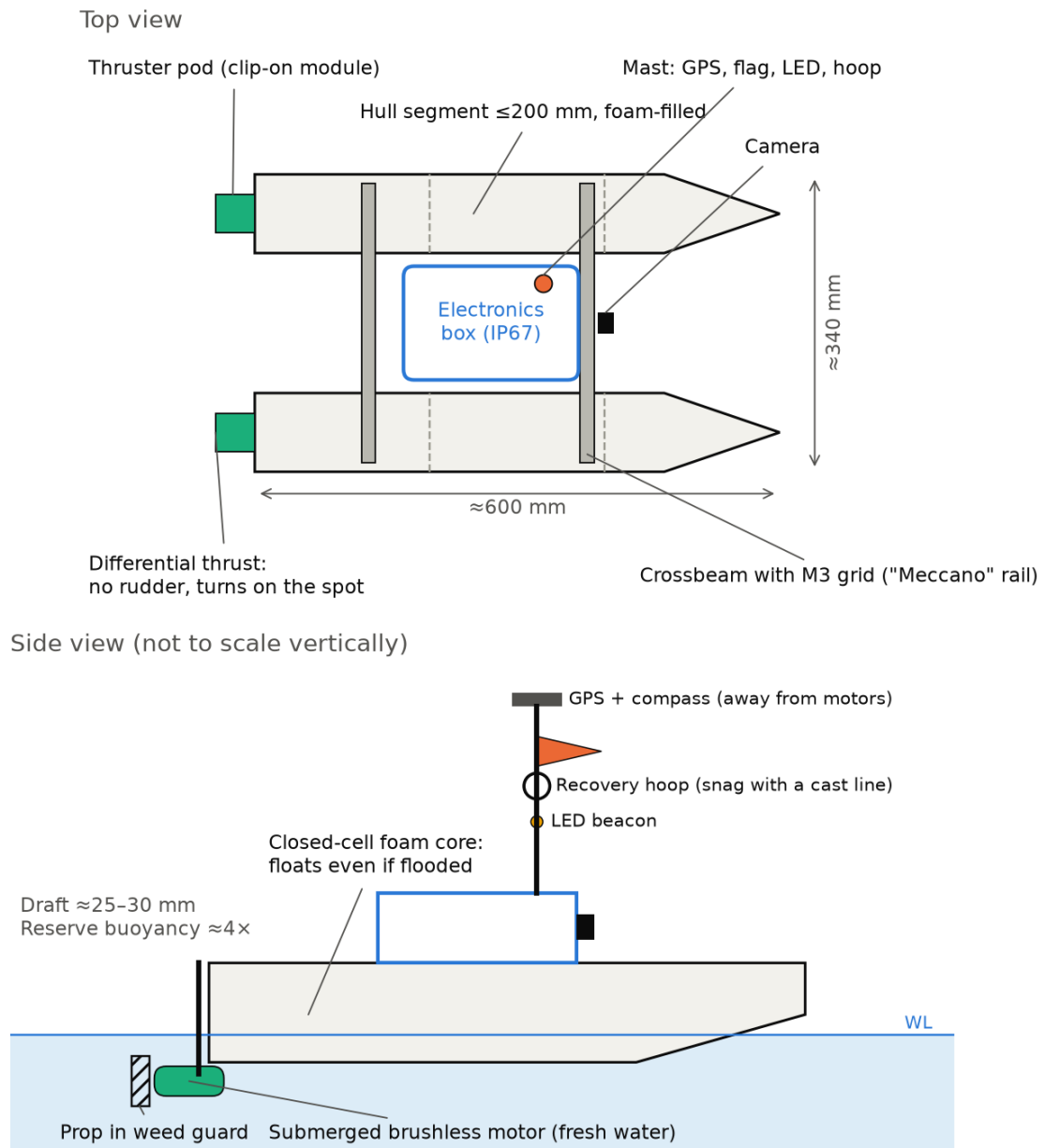


Figure 2. Concept sketch (indicative proportions; detail design follows in the architecture phase).

5.2 Mechanical design and modularity

The build is designed as a kit of modules with standard interfaces, so it can be taken apart, improved and re-combined like Meccano:

Module	Design intent	Interface
Hull segments (×6)	≤200 mm long to fit common print beds. PETG, 3 perimeters. Filled with closed-cell foam so buoyancy never depends on being watertight.	Bolted flanges, M3 heat-set inserts, silicone bead
Crossbeams (×2)	Printed or aluminium box section with a 10 mm M3 hole grid along the top (the 'Meccano rail'). Anything can bolt on anywhere.	M3 clamps to hull bosses
Electronics box	Off-the-shelf clip-lock food box. Cheaper and more reliably sealed than a printed box. All wires go through PG7 glands. A printed tray inside holds the boards.	Printed saddle on the rail
Thruster pods (×2)	Submerged outrunner motor on a printed strut with a printed prop inside a weed guard. Clip on and off for maintenance or a swap to the airboat module.	Dovetail + pin on hull transom; 3-pin waterproof connector
Mast kit	GPS/compass on top, flag, recovery hoop, LED beacon. Printed sockets on a carbon or aluminium tube.	Socket on rail
Payload bay	Free rail space for your son's own modules: a lookout figure, a second camera, a water thermometer, a crane...	M3 grid

Why submerged motors? Brushless outrunners run happily under fresh water; ROV hobbyists do it routinely. That means no shaft seals or stuffing tubes, which are the classic leak point. Rinse and dry after use; the motors are cheap consumables at about £6 each.

5.3 Two brains: clever and safe, kept apart

This is the key architectural idea, carried forward into the next phase. The **safety brain** (ArduPilot on the flight controller) steers the motors and has the final say: fence, failsafes and return-to-launch. The **mission brain** (ESP32-S3) handles Wi-Fi, photos and higher-level behaviour. It talks to the autopilot over MAVLink, the same protocol the ground station uses, so it can request waypoints or a pause but **cannot override the geofence**.

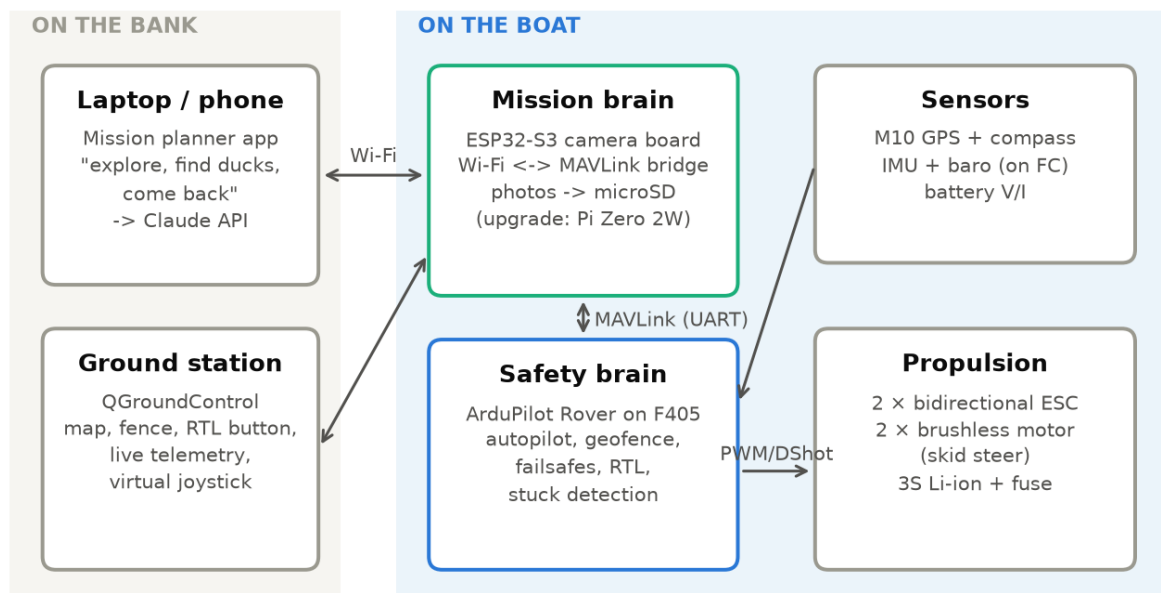


Figure 3. Concept-level block diagram. Detailed interfaces are for the architecture phase.

5.4 Intelligence: from 'drive it' to 'go and find ducks'

The intelligence is built up in levels. Each level is useful on its own and is a milestone to celebrate:

Level	What it does	How
0 Drive	Manual driving from a phone.	QGroundControl virtual joystick over Wi-Fi.
1 Go there	Waypoints, survey patterns, return home.	ArduPilot AUTO missions planned on the map in QGroundControl.
2 Just ask	"Explore the pond, take pictures of some ducks, then come back."	Laptop app sends the request, the pond outline and the fence to Claude. It returns a structured mission (route, photo points, time limit). A validator we write checks it before upload.
3 Captain's log	Picks the duck photos and writes up the voyage.	After return, a vision model sorts the photos, flags the ducks and writes a short log for your son.
4 Spot & look	Notices ducks in real time, pauses and takes a burst of photos, then carries on.	Upgrade: Pi Zero 2W running a small bird detector. Pausing is a request to ArduPilot, still inside the fence.

Design rule: the language model proposes, our validator checks, ArduPilot enforces. The AI never drives the boat directly. A mission is uploaded only if every waypoint is inside the fence, the planned time fits within half the battery, the photo count is sensible, and the mission ends in return-to-launch. Your son sees the plan on the map and presses GO.

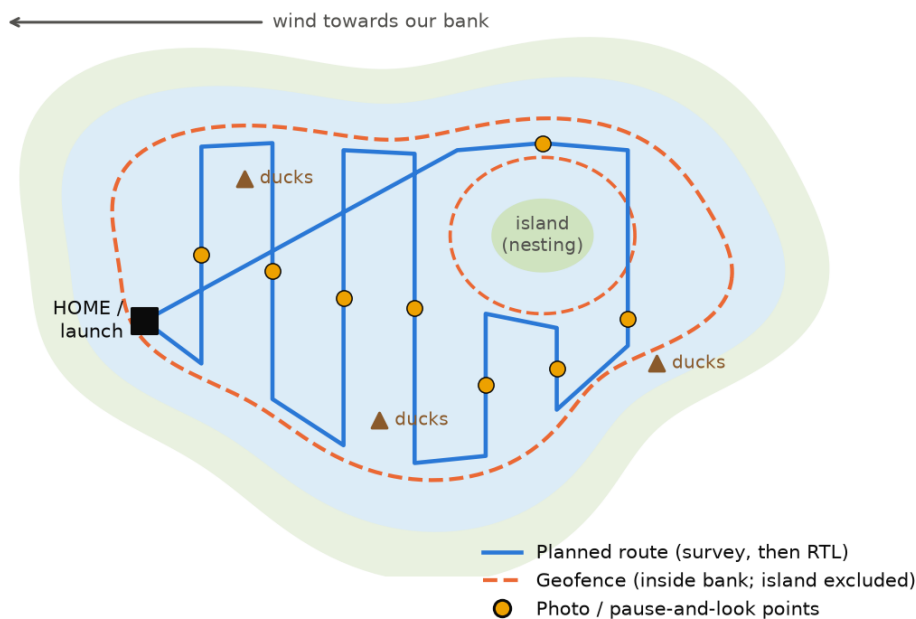


Figure 4. What "explore this pond, take some pictures of ducks, then come back" might become: a survey route inside the fence, photo points, and RTL. The nesting island is fenced out. We don't chase wildlife.

5.5 Can't-lose-it strategy

Reliability is designed in as independent layers. Any one layer can fail and the next still saves the boat:

1 Keep it afloat	Foam-filled hulls float even if flooded · catamaran is very hard to capsize
2 Keep it inside	ArduPilot polygon geofence 5 m inside the bank · islands & reeds excluded
3 Get it back	Return-to-Launch on fence breach, low battery, lost link, mission end; Hold if stuck
4 Find it	Hi-vis hull, flag, LED beacon · last GPS fix in QGC · optional Bluetooth tag
5 Operate smart	Wind blowing TOWARDS our bank · pre-launch checklist · recovery kit ready

Each layer works on its own. Layers 1-3 live on the boat and need no radio link.

Failure	What the boat does	ArduPilot feature
Drives towards the bank / off the map	Stops at the fence, returns home	Polygon fence, action = RTL
Battery getting low	Returns home with $\geq 30\%$ left	Battery failsafe (voltage and mAh used)
Wi-Fi link lost	Finishes the leg, then returns home	GCS failsafe, action = RTL
GPS or compass goes bad	Stops motors and holds	EKF failsafe, action = Hold
Wrapped in weed, not moving	Stops (optionally reverses to shed weed), alerts us	Crash/stuck check + Lua script
Mission-brain crash	Autopilot carries on the mission and returns home	Mission lives on the flight controller
Everything dead	Floats, drifts downwind to our bank, beacon flashing	Physics + choosing the launch bank

Recovery kit on the bank: a telescopic pole with a landing net for near-bank rescues, and a spinning rod with a weighted float line cast over the mast hoop for anything further out. First autonomous trials happen in a small pond where every point is within casting range.

5.6 Budget

Core Mk1 build: everything needed for Levels 0-3 of the intelligence ladder. The estimate lands at ≈£105, within about 5% of the £100 target. The software (ArduPilot, QGroundControl, Python) is free. The Claude API costs a few pence per mission plan.

#	Item	£ (approx.)
1	Flight controller, F405-class with ArduPilot firmware support (e.g. SpeedyBee F405 Mini / Matek class - confirm on ArduPilot's board list)	25
2	GPS + compass module, u-blox M10 class	14
3	Seeed XIAO ESP32-S3 Sense (camera, microSD slot, Wi-Fi with external antenna)	14
4	microSD card, 32 GB	4
5	2 × brushless outrunner motor, 2204-2208 size (runs submerged in fresh water)	12
6	2 × bidirectional ESC, 20-30 A (BLHeli_S / AM32 '3D mode')	10
7	Battery: 3S Li-ion pack (3 × 18650 + BMS) or 3S 2200 mAh LiPo	15
8	IP67 clip-lock food box + 4 × PG7 cable glands + inline blade fuse	6
9	Closed-cell foam (XPS offcut / pipe lagging), hi-vis tape, LED beacon	5
Core cash spend		≈105
...if you already own a 3S battery		≈90
Workshop stock (not counted): PETG filament (~0.6 kg), M3 fasteners & heat-set inserts, wire, - XT30 connectors, solder, silicone sealant, 3S balance charger		

Optional upgrades (later, when Mk1 has earned them)

Upgrade	£	What it adds
Raspberry Pi Zero 2W + Camera Module 3	£40	On-board duck detection, 'pause and look' behaviour, much better photos.
ExpressLRS handheld transmitter + receiver	£55-70	Independent second link for manual override; longer range than Wi-Fi.
Bluetooth item tracker (AirTag / SmartTag) inside the box	£25-30	Belt-and-braces 'where is it' if everything else fails.
Airboat thruster-pod module (2 × 2205 + 5" props + printed guards)	£10	Swap-in module for weedy ponds. A fun second propulsion experiment.
Cost-down levers if needed: a cheaper F405 clone (-£8); brushed motors with a dual brushed ESC (-£6, less efficient); salvaged 18650 cells (-£8, test capacity first).		

5.7 Sanity-check numbers

Check	Working	Result
Mass	Hulls + foam 500 g, box + electronics 250 g, battery 150 g, pods 150 g, beams + mast 200 g, misc 150 g	≈1.4 kg
Draft	1.4 L displaced over 2 hulls ≈ 600 × 90 mm waterplane, Cwp≈0.7	≈25-30 mm
Reserve buoyancy	Hull volume ≈2 × 600×90×100 mm × 0.6 ≈ 6.5 L vs 1.4 L needed	≈4.5×
Endurance	3S 18650 ≈33 Wh, 80% usable ≈26 Wh; cruise ≈8 W propulsion + ≈2.5 W electronics	≈2.5 h theoretical; claim ≥60 min
Mission length	100 × 60 m pond, 10 m lane spacing ≈ 600 m at 1 m/s	≈10 min + photo pauses
Position accuracy	M10 GPS typically 1.5-3 m	Fence 5 m inside bank
All to be verified in the bath, then the paddling pool, then the pond. Early test data feeds the architecture phase.		

6. How your son gets involved: build roadmap

Each phase ends with something that works and a reason to celebrate. The simulator phase means the software can be played with at the kitchen table before any plastic is printed.

Phase	What we do	Son's role	Done when
0 Simulator	Run ArduPilot SITL + QGroundControl on the laptop, using the real pond from satellite imagery.	Draws the fence, plans missions, 'crashes' the virtual boat on purpose to watch failsafes fire.	Virtual boat completes a mission and returns home.
1 Hulls	Print hull segments, foam-fill, bath float test, measure draft and add ballast.	Archimedes experiment: predict the draft, then measure it. Paints the livery.	Floats level at predicted draft with test weights.
2 Power & drive	Thruster pods, ESCs, battery, flight controller. Driven manually from the phone.	Assembles pods, tests clip-on/clip-off, first drive in a paddling pool.	Drives, turns on the spot, reverses.
3 Autonomy	GPS, compass calibration, fence, failsafes. Small-pond trials with deliberate failsafe tests.	Mission commander: runs the pre-launch checklist.	Every failsafe in section 5.5 demonstrated on the water.
4 Camera & AI	ESP32 photo capture, natural-language mission app, captain's log.	Gives the orders, reviews photos, names the ducks.	"Explore, photograph ducks, come back" works end to end.
5 His module	Open-ended: he designs a module for the rail.	Designer: sketch, Tinkercad, print, fit.	His module sails.

7. Risks and mitigations

Risk	Likelihood / impact	Mitigation
Weed fouls the props	High / medium	Weed guards, reverse-to-clear routine, stuck detection → Hold, airboat module for weedy ponds, and survey the pond first.
Water gets into the electronics	Medium / medium	Sealed food box + glands, silica gel, and a 'dunk test' before every season. Foam hulls mean flooding can't sink it.
Compass interference → poor steering	Medium / low	GPS/compass up the mast, twisted power leads, calibrate on site, compare against GPS course.
Budget creep	Medium / low	Core BOM frozen for Mk1. Upgrades only after Phase 4.
Wi-Fi range shorter than hoped	Medium / low	Safety doesn't depend on it. External antenna, ELRS upgrade available.
Wind stronger than the boat	Low / medium	Wind limit in the checklist (no trips above a light breeze), and launch from the downwind bank.
Lithium battery fault	Low / high	BMS or LiPo checker, inline fuse, charge in a LiPo bag, never charge unattended, and store at storage voltage.

8. Rules, wildlife and good manners

- **Permission.** Many park and council ponds have byelaws on model boats, and some have a model-boat club with set times. Check who owns the water and ask first. A friendly local club is also a great place for trials.
- **Wildlife.** The Wildlife and Countryside Act 1981 protects wild birds and their nests. The mission design never chases birds, keeps well clear of islands and reed beds (fenced out), and avoids approaching nesting areas in the breeding season (roughly March to August). Photos are taken from a respectful distance. Ducks that swim up to the boat on their own are a bonus.
- **Radio.** 2.4 GHz Wi-Fi and ExpressLRS are licence-exempt in the UK at normal power levels. Avoid cheap 433 MHz 'telemetry radios' sold at powers above UK limits.
- **People.** Nobody goes into the water to fetch the boat. Adult supervision at the bank at all times.

9. Decisions and inputs needed before architecture design

To make the architecture concrete, it would help to know:

#	Question	Why it matters
1	Which pond(s)? Rough size, weed, islands, launch points, and whether model boats are allowed.	Sets fence geometry, range needs, propulsion choice (submerged vs airboat pod).
2	Your son's age and what he's most excited by (building, driving, coding, photos?).	Tunes the module split and which parts he designs himself.
3	Existing kit: 3S battery and charger, RC transmitter, Raspberry Pi, print bed size?	Changes the budget and the link design.
4	OK to use the Claude API from a laptop for mission planning (needs internet on the bank, e.g. phone hotspot), or prefer offline planning at home?	Decides the ground-side software architecture.
5	Happy with ArduPilot as the safety core?	It's the backbone of the reliability argument. The alternative is concept C.

Next step: architecture design. Detailed electrical schematic and power budget; MAVLink interfaces between the brains; ArduPilot parameter set (fence, failsafes, skid-steer tuning); CAD layout and hull lines; the mission-planner app and its validator; and a test plan for each failsafe.